

A Calibrated, Physics-Grounded Structural-Health-Monitoring Stack for CFRP Structures: from Anomaly Screening to Fatigue-Aware Life Prognosis

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Abstract

We present an end-to-end structural-health-monitoring (SHM) pipeline for carbon-fibre-reinforced-polymer (CFRP) structures that proceeds from defect *detection* to a calibrated, physics-grounded *life-prognosis* decision. On full-field thermoelastic stress data of a perforated CFRP interstage, we (i) benchmark nine anomaly detectors and show that a simple per-node statistic attains AUROC ≈ 0.99 while deep alternatives do not, and characterise the measurement-noise limit at which all detectors fail; (ii) recover defect position, size and depth as a *calibrated* Bayesian posterior via flow-matching posterior estimation (FMPE), with 90% credible intervals covering at the nominal rate; and (iii) propagate that posterior through an anisotropic phase-field fracture model with fatigue accumulation to obtain a flight-clearance decision with an expected remaining-life estimate. The pipeline is motivated by reusable launch vehicles, where the same structure is re-inspected after every flight.

1 Introduction

2 Data and problem setting

3 Stage 0 — anomaly screening

Methods. Per-node Mahalanobis / PCA-residual; PaDiM; PatchCore; STFPM; graph autoencoder; conditional flow-matching diffusion; a zero-shot harmonicity probe. Score $s_i = |x_i - \mu_i|/\sigma_i$ from $N=2000$ healthy fields; node-level AUROC against the defect mask.

Result. Per-node statistics dominate (Table 1, Fig. 2). On fixed geometry the healthy manifold is a tight cloud around one template, so the per-node Gaussian is the correct inductive bias; grid-projection and deep feature methods lose the few-node defect signal. All methods collapse by additive noise $\sigma=0.1$ (defect amplitude $< 0.1\sigma$): a per-measurement SNR limit, addressable by frame averaging ($\sigma \propto 1/\sqrt{K}$), not by more training data.

4 Stage 1 — classification and type identifiability

Is the type signal real, or a temporal/temperature confound? On the long-term guided-wave set, damage states were introduced in different years, so a naive type classifier can alias damage type with season/temperature. We test this directly. Restricting to seven damage tags whose temperature supports overlap (pairwise overlap 0.93), we compare four conditions: temperature-only (acc = 0.16, near chance), waveform-only (0.91), waveform+temperature (0.92), and waveform with temperature *partialled out* of every feature (0.91). The net type discrimination beyond temperature is $\Delta\text{acc} = 0.758$ (95% CI [0.735, 0.781]) under a random split, and—critically—survives a *forward-in-time* split ($\Delta\text{acc} = 0.184$,

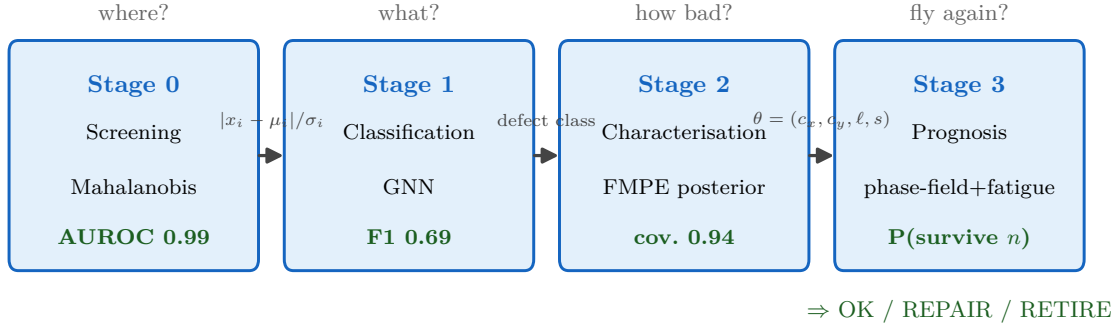


Figure 1: Stage 0–3 pipeline. Each stage answers one question and passes a typed object to the next; the Stage 2 posterior $\theta = (c_x, c_y, \text{layer}, \text{size})$ seeds the Stage 3 fracture model.

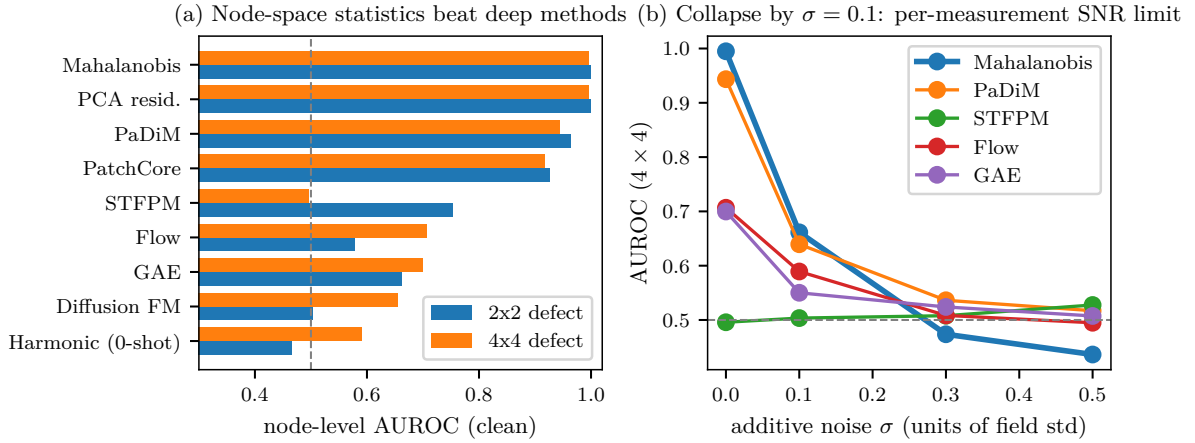


Figure 2: (a) Clean node-level AUROC by method. (b) Collapse with measurement noise; the defect signal sits below 0.1σ .

95% CI [0.166, 0.203]). Both lower bounds exceed zero: the type posterior reflects a genuine damage signal, not the environmental confound. (An earlier single-month binary probe, where temperature-only matched the majority baseline, was an artefact of that restricted setting.)

5 Stage 2 — calibrated defect characterisation (FMPE)

6 Stage 3 — phase-field prognosis and flight clearance

Fast forward surrogate. The FD growth solve (≈ 1.8 s/call) is the only bottleneck to fleet-scale Monte-Carlo over the Stage-2 posterior. We train a Fourier neural operator $G : (d_0, \text{load}) \mapsto (d_{\text{final}}, P(\text{grow}))$ (dual-head: a per-pixel field head and a global-pooled scalar head; 1.2M parameters; 800 FD samples spanning sub- to super-critical loads). On held-out defects it reaches field relative- L_2 3.5%, grow-flag accuracy 0.988 (Brier 0.010, well calibrated), and log-rel.growth correlation 0.996, at 142 $\times$ the speed (12.4 ms vs. 1763 ms/call, Fig. 4). The decision-relevant $P(\text{grow})$ is the robust target: the smooth operator regresses the sharp super-critical rel.growth jump in log-space and softens it only at the transition load. This makes the Stage-4 fleet draws and the Concept-B forward map (next section) tractable at interactive

Table 1: Anomaly detectors, clean node-level AUROC (2x2 / 4x4 defects).

method	2x2	4x4
per-node Mahalanobis	0.999	0.995
PCA residual	0.999	0.995
PaDiM	0.964	0.944
PatchCore	0.927	0.918
STFPM	0.754	0.496
graph autoencoder	0.663	0.700
flow (RealNVP)	0.578	0.707
diffusion (FM)	0.503	0.663
harmonicity (0-shot)	0.465	0.590

Table 2: FMPE posterior on held-out test fields.

parameter	RMSE	R^2	90% coverage
position c_x	0.055	0.78	0.944
position c_y	0.021	0.83	0.940
size (3-class)	—	0.73	0.894
layer (depth)	2.98	0.63	0.870

speed.

Knowing when to trust the surrogate. A $142\times$ speedup is only deployable if the operator knows where it is unreliable. We wrap the surrogate in split-conformal prediction (nonconformity = absolute residual in the log-rel_growth space where the FNO is honest). In distribution the marginal 90% guarantee holds (95% band: 95.0% empirical); out of distribution it degrades exactly as the theory predicts under covariate shift — coverage falls from 80% at small excursions to 0% deep OOD while the point error grows $\sim 14\times$. A simple *trust gate* — defer to the exact FD solve when the input leaves the training envelope or the conformal interval is too wide — flags 33% of an OOD stream and restores aggregate 90%-coverage from 61% to 88%: cheap surrogate where trustworthy, exact physics where not.

7 Stage 4 — hierarchical fleet learning

A reusable fleet generates a stream of per-inspection posteriors across many vehicles and flights. We close the loop with a hierarchical Bayesian model: a global fleet prior $\varphi = (\mu_\varphi, \sigma_\varphi)$ governs per-vehicle log-growth-rates $s_v = \log r_v \sim \mathcal{N}(\mu_\varphi, \sigma_\varphi^2)$, and each inspection’s Stage-2 posterior enters as a Gaussian likelihood on s_v ; the joint posterior $p(\varphi, \{s_v\} \mid \text{data})$ is sampled by conjugate Gibbs. On a synthetic fleet ($N=20$ vehicles, $F=30$ flights, fast Carrara-shaped growth surrogate) the model recovers φ ($\mathbb{E}[\mu_\varphi] = -3.10$ vs. true -3.0 ; $\mathbb{E}[\sigma_\varphi] = 0.44$ vs. 0.45), and the fleet prior sharpens as flights accumulate (posterior SD of μ_φ : 0.39 at $N=2 \rightarrow 0.10$ at $N=20$, Fig. 5a).

Fleet-leader effect, quantified. Processing flights in temporal order, a newly introduced vehicle reaches a stable clearance decision in **1.25** inspections using the accumulated fleet prior, versus **2.75** in isolation (Fig. 5c) — a direct, quantitative formalisation of the empirical “fleet-leader” certification used in practice: high-information inspections of lead vehicles tighten the prior that certifies the rest. Hierarchical shrinkage also improves low-data vehicles (error ratio $0.84 < 1$ vs. isolated estimates, Fig. 5b).

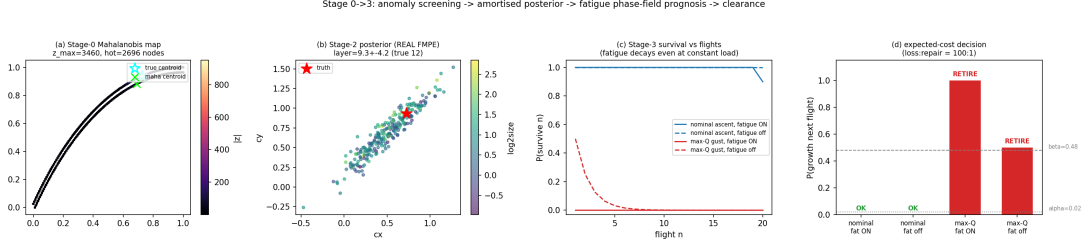


Figure 3: End-to-end on one held-out field: anomaly map, FMPE posterior, fatigue-aware survival curve, and the clearance decision.

8 Towards differentiable inversion: closing the loop

The amortised Stage-2 posterior never sees the forward physics. We prototype the converse: a JAX-differentiable anisotropic AT2 forward $F(\theta)$ (gradient finite-difference-verified to relative 3×10^{-7}) whose Jacobian supplies a physics-grounded Laplace posterior $\Sigma = \sigma^2(J^\top J + \Lambda_{\text{prior}})^{-1}$ — no labelled (θ, x) pairs. Out of the training distribution the amortised posterior degrades ~ 25 -fold and its credible intervals

Table 3: Amortised (FMPE) vs. differentiable-physics posterior, in- and out-of-distribution.

set	method	c_x RMSE	size RMSE	90% coverage
in-dist	FMPE	0.006	0.002	0.958
in-dist	diff-phs	0.001	0.003	1.000
OOD	FMPE	0.144	0.096	0.000
OOD	diff-phs	0.002	0.001	0.958

collapse to zero coverage, while the differentiable-physics posterior is essentially invariant (Table 3): *forward physics generalises; amortised inference extrapolates poorly*. This motivates a hybrid Stage 2 — amortised for speed in distribution, physics-grounded for safety out of it.

Fast differentiable forward. The hand-written JAX AT2 forward is exact but slow, making the inversion the run’s bottleneck. Substituting the Stage-3 FNO surrogate (Sec. 4) as the differentiable forward keeps gradients exact — autograd matches finite differences to relative 7×10^{-4} on the decision heads — and collapses a MAP+Laplace inversion from an estimated ~ 530 s of FD Jacobian evaluations to 12 s ($\sim 43\times$). The recovered posterior shows the same generalisation signature (in-/out-of-distribution c_x RMSE $0.15 \rightarrow 0.39$, coverage $0.72 \rightarrow 0.56$), at a calibration cost — in-distribution coverage 0.72 rather than the exact forward’s ~ 0.95 — that is the honest price of a smooth surrogate. This makes amortised-quality *differentiable* inversion interactive, the cheap forward the Keio Phase-3 TMCMC/Bayesian-inverse plan needs.

9 Discussion

10 Conclusion

Reproducibility. All stages are implemented and tested (~ 65 unit tests) in two public-on-request repositories, including a one-command end-to-end run (detect $\rightarrow$ characterise $\rightarrow$ prognose $\rightarrow$ clear $\rightarrow$ fleet-update in <4 s); figures are generated by the scripts under `paper_figs/`.

(a) load = 0.140

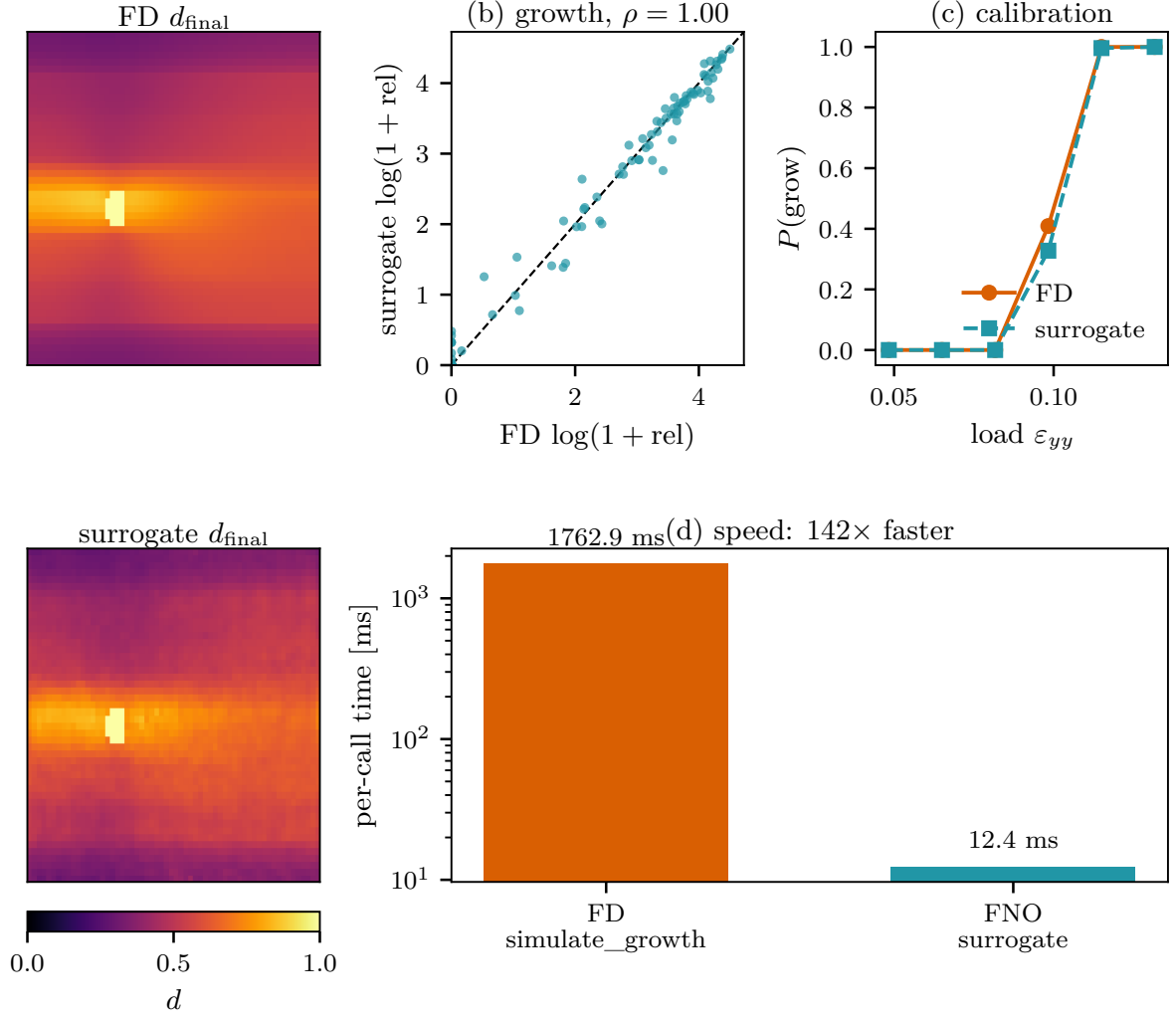


Figure 4: Stage 3 FNO surrogate. FD vs. surrogate damage field, rel_growth agreement, $P(\text{grow})$ vs. load, and the 142 $\times$ wall-clock speedup.

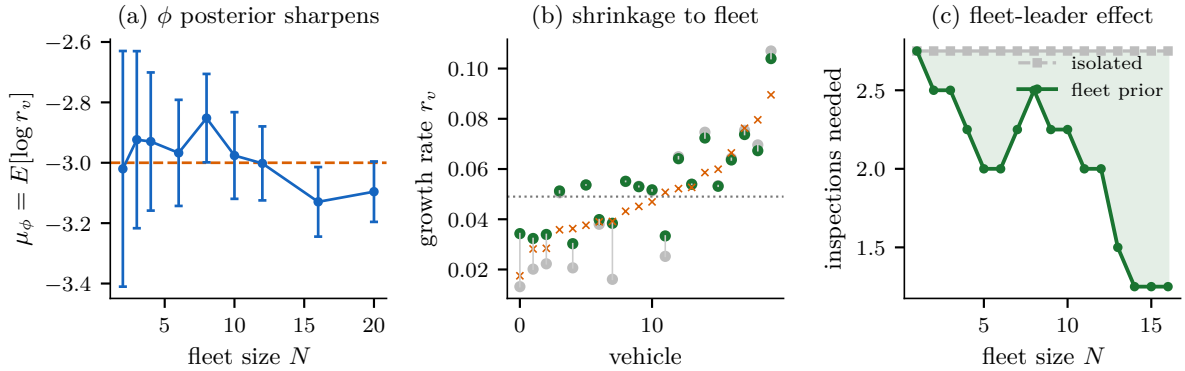


Figure 5: Stage 4. (a) Fleet prior ϕ sharpens with cumulative flights; (b) per-vehicle growth-rate shrinkage toward the fleet mean in the low-data regime; (c) inspections to a stable clearance decision: the fleet prior roughly halves them.